Viacheslav Kruglov

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Profile

C++ software engineer with a PhD in Physics and several years of experience developing technical software under Linux. Experience in C++, Python, simulation, data processing, debugging, performance optimisation, and maintainable codebases. I am looking for a role in C++ development, technical software, simulation software, or performance-critical systems.

Core Skills

Languages: C++ (C++17/20), Python

Software Development: Linux, Bash, Git, CMake, GDB, maintainable code, debugging, troubleshooting

Technical Focus: simulation, data processing, debugging, performance

Parallelisation: Intel TBB, OpenMP, SLURM

Numerics: interpolation, filtering and smoothing, FFT usage, applied linear algebra

Libraries and Formats: LAPACK, FFTW, NetCDF, NumPy, pandas

Selected Project

particle-transport-framework [GitHub]: C++ framework for large-scale particle transport simulation and ocean data processing on Linux. Designed for long-running simulations on gridded flow fields. Includes NetCDF and HYCOM handling, kd-tree spatial queries, interpolation, local coordinate transformations, and parallel processing with Intel TBB.

Professional Experience

2023–2026: **PostDoc / Research Software Developer**, University of Oldenburg, Oldenburg

- Developed and maintained a C++ codebase for large-scale time-dependent and multi-dimensional data processing under Linux.
- Worked on simulation workflows with millions of tracers and more than 10 billion simulated trajectories per model year.
- Implemented numerical components, data structures, coordinate transformations, and analysis routines in C++ and Python.
- Reduced runtime and memory use through profiling, data-structure changes, and parallelisation with Intel TBB.
- Set up reproducible batch runs for simulation and analysis, including automated evaluation and SLURM-based execution.
- Debugged logical and numerical issues to improve correctness and reproducibility.

2017–2022: **Research Scientist (Computational Physics)**, Kotelnikov Institute of Radioengineering and Electronics, RAS (Saratov Branch), Saratov

- Developed C++ software for numerical experiments in nonlinear dynamical systems, including long-running parameter studies and automated sweeps.
- Executed multi-parameter sweeps of roughly 10^6 parameter combinations and processed thousands of long-duration trajectories.
- Used LAPACK and FFTW in computational workflows and implemented custom analysis code in C++ and Python.

- Worked on reproducible runs and on diagnosing numerically unstable model behaviour.

2016–2017: **Software Engineer (Industrial Simulation)**, INACOS (INIUS JSC), Saratov

- Implemented thermodynamic property models in industrial simulation software.
- Integrated a vacuum distillation column model into an existing simulation platform.
- Compared simulation outputs against references and corrected model and implementation issues.

Education

Physics and Mathematics **PhD**
Saratov State University

Physics **Diploma**
Saratov State University

Additional Information

Languages: English (fluent), German (B1 certified, currently improving toward B2), Russian (native)

Work authorization: Permanent residence permit in Germany, no sponsorship required